Charles M. Bridges, M.S., Ph.D.

Microbiologist | Molecular Biologist | Biochemist | Scientific Consultant

charlesmbridgesphd@gmail.com | LinkedIn.com/in/charlesmbridges | charlesmbridges.github.io/professional

Professional Summary

Biotech-focused scientist with broad experience in microbiology, molecular biology, genetics, genomics, and biochemistry. Strong track record of generating reliable, reproducible data, improving laboratory workflows, generating open-source resources and contributing to publication-quality research. Experienced working in BSL-2 laboratory settings, troubleshooting complex technical workflows, and communicating scientific results to technical and non-technical audiences. Recognized for improving operational efficiency through thoughtful problem-solving, practical risk management, and effective training and mentorship.

Areas of Expertise

- **Laboratory & Scientific Operations**: BSL-2 Laboratory Operations | Procurement & Purchasing | Inventory Management | Budget Support | Account Allocation | Documentation & Compliance | Vendor Coordination | Laboratory Information Management Software (LIMS) | Quality Control | SOP Development
- **Experimental Techniques**: Sterile Technique | Microbial & Viral Culture (Aerobic & Anaerobic) | Nucleic Acid Extraction | Molecular Cloning | Mutagenesis | Microbe Engineering | CRISPRi | Protein Engineering, Expression & Purification | Liquid Chromatography (LC) | Ultracentrifugation | Fluorescence & Transmission Electron Microscopy
- **Technical, Computational & Bioinformatics**: Unix | Linux | HPC | Bash | Python | R | High-Throughput Sequencing & Quality Control | Genome Assembly & Annotation | Comparative Genomics | Data Visualization | Scientific Reporting
- **Leadership & Compliance**: Laboratory Management | Preventive Maintenance | Training & Onboarding | EHS & DPH Coordination | Inspection Readiness | Granting Agency Compliance | Risk Assessment
- **Communication & Collaboration**: Cross-functional Teamwork | Collaborative Communication | Scientific Writing | Technical Presentations | Electronic Recordkeeping | Mentoring | Problem-solving | Interdisciplinary Thinking

Professional Experience

Research Associate 1 / Laboratory Manager, University of Connecticut, Storrs, CT | December 2025 – Present

- Manage day-to-day operations for a BSL-2 virology and biochemistry laboratory, ensuring continuous research execution, safety compliance, and inspection readiness.
- Design and execute virology and biochemistry experiments; analyze data and support manuscript preparation and publication deliverables.
- Manage laboratory purchasing and supply logistics, including requisitions, inventory tracking, purchase reconciliation, and allocation of expenses to appropriate funding accounts.
- Lead equipment procurement, onboarding, training and service scheduling, minimizing downtime and protecting experimental continuity.
- Provide strategic oversight for interdisciplinary, cross-functional research projects by aligning timelines, experimental priorities, and deliverables across collaborators.
- Ensure University Environmental Health & Safety (EHS) and CT Department of Public Health (DPH) laboratory inspection readiness, including supporting documentation, communication, and operational follow-through.

Founder & Principal Consultant, Strategic Molecular Solutions LLC, Granby, CT | March 2025 – Present

- Provide consulting across molecular biology, microbiology, and biotechnology strategy for academic and industry clients, including troubleshooting, experimental planning, and workflow optimization.
- Translate research goals into actionable experimental plans with practical constraints (budget, feasibility, timelines, regulatory context).

Large Language Model (LLM) Subject Matter Expert, Handshake AI, Remote | August 2025 – December 2025

- Created, evaluated and refined multimodal artificial intelligence (AI) Large Language Model (LLM) prompts, responses and rubric quality, supporting improved model reliability and instruction-following.

Adjunct Faculty Instructor of Virology, University of Connecticut, Storrs, CT | January 2024 – May 2024

- Delivered curriculum for a combined graduate/undergraduate virology course for 24 graduate and undergraduate students, including projects, presentations, lectures, and student assessments.

Postdoctoral Research Associate, University of Connecticut, Storrs, CT | July 2021 – Dec. 2025

- Led experimental execution and research strategy for cross-functional virology and biochemistry projects, delivering publishable datasets under project deadlines.
- Drove projects from concept through experimental design, *in silico* modeling, troubleshooting, statistical interpretation, and manuscript development.
- Improved laboratory productivity by identifying and mitigating recurring bottlenecks such as sample processing, experimental failure points and protocol ambiguity.
- Contributed large-dataset genomics analysis to 2 peer-reviewed publications in mBio, supporting cross-institutional collaborations.
- Designed over 200 oligonucleotides for use in developing over 145 plasmid constructs and over 300 bacterial strains for use in protein expression, CRISPRi, virus-like particle (VLP) production and viral mutational analysis.

Graduate Assistant, University of Connecticut, Storrs, CT | August 2011 – August 2021

- Performed molecular microbiology research resulting in 5 peer-reviewed publications in genomics, microbial physiology, genetics and synthetic biology.
- Designed over 300 oligonucleotides and engineered over 80 plasmid constructs and over 230 bacterial strains for *in vitro* molecular, physiological, microfluidic assays and *in vivo* studies in plants and insects.
- Conducted bacterial isolation of over 85 bacterial strains and performed cultivation, genetic & physiological screens and molecular workflows, producing reproducible datasets used in publications.
- Conducted comparative genomics analyses to identify and interpret genomic traits across 4 novel bacterial isolates, supporting mechanistic research conclusions that lead to 3 peer-reviewed publications.
- Supported undergraduate laboratory programs through instruction, oversight, and evaluation in a variety of microbiology and bacterial genetics courses for over 500 students.

Education

Doctor of Philosophy, Molecular and Cell Biology, University of Connecticut, Storrs
Master of Science, Microbiology, University of Connecticut, Storrs
Bachelor of Science, Molecular and Cell Biology, University of Connecticut, Storrs

Advanced Training

National Center for CryoEM Access and Training (NCCAT), New York, NY | April 2025

- Cryo-electron microscopy (cryo-EM) sample & grid preparation, computational dataset workflow

Bacterial and Viral Bioinformatics Resource Center (BV-BRC), Argonne National Laboratory | February 2025

- Pathogen identification, genomic virulence analysis, epidemiological outbreak tracking and analysis

Selected Publications

1. Juliana R. Cortines, **Charles M. Bridges** et. al. (2024). Transition metals and oxidation reactions trigger stargate opening during the initial stages of the replicative cycle of the giant Tupanvirus. mBio Virology.
2. Justin C. Leavitt, [..], **Charles M. Bridges**, et. al. (2023). Bacteriophage P22 SieA-mediated superinfection exclusion. mBio Bacteriophages.
3. Christopher J. Hawxhurst*, Jamie L. Micciulla*, **Charles M. Bridges**, et. al. (2023). Soil protists can actively redistribute beneficial bacteria along *Medicago truncatula* roots. Applied and Environmental Microbiology.
4. **Charles M. Bridges** & Daniel J. Gage. (2021). Draft genome sequences of *Dysgonomonas* sp. strains GY75 and GY617, isolated from the hindgut of *Reticulitermes flavipes*. Microbial Resource Announcements.
5. **Charles M. Bridges** & Daniel J. Gage. (2021). Development and application of aerobic, chemically defined media for *Dysgonomona*s. Anaerobe.
6. Azady Pirhanov, **Charles M. Bridges**, et. al. (2021). Optogenetics in *Sinorhizobium meliloti* enables spatial control of exopolysaccharide production and biofilm structure. ACS Synthetic Biology.

Selected Presentations

Selected Speaker, XXVIII Biennial Conference on Phage/Virus Assembly | June 2023
Selected Speaker, Pioneer Valley Microbiology Symposium | January 2020
Invited Speaker, Natural Science Deptartment Science Colloquium, Castleton University | October 2019

Selected Awards

UConn Connects David T. Ouimette Outstanding Mentor Award | September 2023
Best Talk, Pioneer Valley Microbiology Symposium | January 2020